AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

Biology Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are ELEVEN questions. Answer ALL questions. Questions 10 and 11 offer TWO choices. Attempt any ONE choice from each.

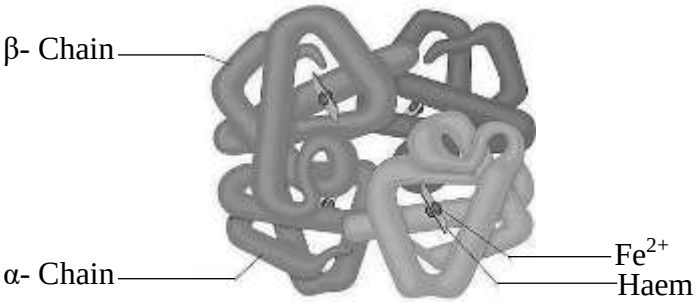
3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. The given diagram shows the quaternary structure of a protein (haemoglobin). (2 Marks)



Why is this protein identified as a quaternary structure? Identify any ONE type of interaction (bond) found between α-chain and β-chain.

b. Two proteins with the same number and type of amino acids may have different tertiary structures.

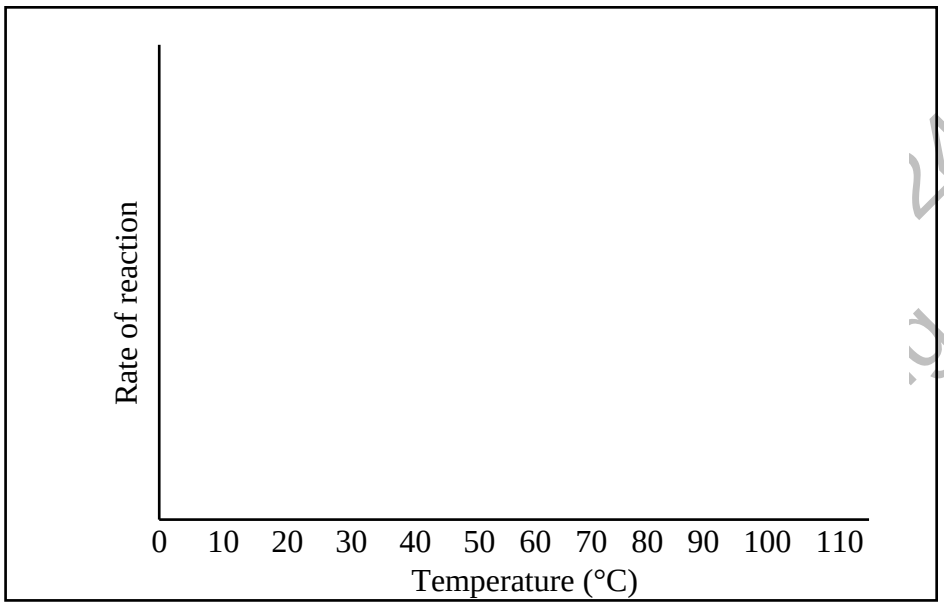
Support the given statement by giving TWO reasons. (2 Marks)

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Q.2. (Total 2 Marks)

Draw TWO generalised curves on the given template to show the effect of temperature on the rate of enzyme-catalysed reaction for each of the following:

- A typical human enzyme
- A thermophilic bacterial enzyme



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Q.3. (Total 2 Marks)

Mention any ONE function of each of the following parts of cytoskeleton in eukaryotic cells:

a. Microtubules (1 Mark)

_____

_____

b. Microfilaments (1 Mark)

_____

_____

Q.4. (Total 2 Marks)

Write any ONE example of each class of phylum mollusca in the given table.

Class	Example
Gastropoda	
Cephalopoda	

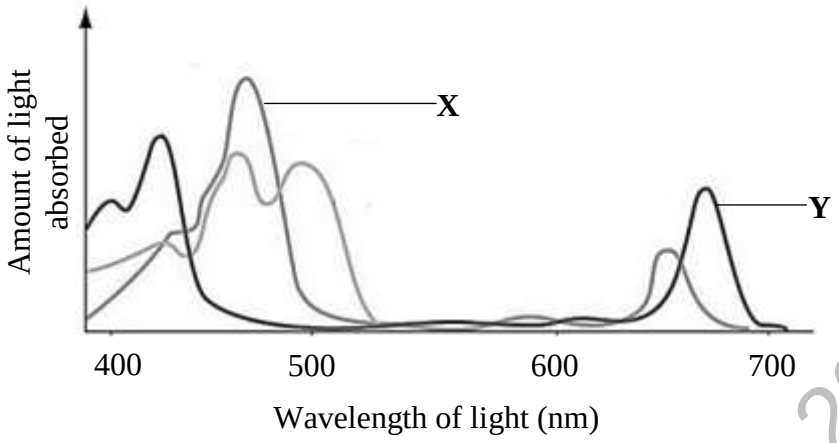
Q.5. (Total 3 Marks)

Describe any THREE adaptive features of seeds that assist the survival of higher plants on dry land.

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Q.6. (Total 2 Marks)

The given graph shows the absorption spectrum of two different pigments **X** and **Y**.



Based on the wavelength of light absorbed, identify pigments **X** and **Y**.

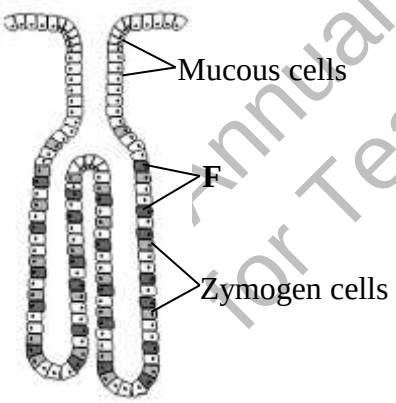
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Q.7. (Total 2 Marks)

The given diagram shows gastric glands in the stomach wall of a human being.



Identify the cells labelled as **F**. How do these cells help in the process of digestion?

_____

_____

_____

_____

Q.8. (Total 2 Marks)

When a person suffering from emphysema smokes, the bronchioles and alveoli get damaged.

Describe how the damage of each structure will make the breathing harder in this person.

_____

_____

_____

_____

Q.9. (Total 2 Marks)

A doctor listening to the heartbeat through a stethoscope hears two sounds, *lub* and *dub*.

How are these two sounds produced?

_____

_____

_____

_____

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Q.10 (Total 7 Marks)

EITHER

a. Human pancreatic cells secrete trypsinogen enzyme.

Describe the role of each of the given organelles in the synthesis and secretion of trypsinogen.

(7 Marks)

Nucleus	Ribosomes	Secretory vesicles
Golgi apparatus	Nucleolus	Endoplasmic reticulum

(**Note**: Your answer should be in the proper order of occurrence.)

OR

b. According to an article published by Nic Fleming on February10, 2015, "The most dominating form of life, in terms of their number on the Earth, is six-legged, shrimp-like springtails … The 6,000 known species of these wingless arthropods can be found in all manner of habitats all over the world, from beaches and cliffs to the Antarctic and the highest mountain ranges on the Earth."

i. Explain any FOUR ways in which the organisms described in the given text have successfully adapted to diverse habitats. (4 Marks)

ii. Describe any THREE ways in which these organisms impact our economy. (3 Marks)

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Q.11. (Total 7 Marks)

EITHER

a.

i. How does gaseous exchange take place in hydra? (2 Marks)

ii. Describe any FIVE general properties of respiratory surfaces in higher animals. (5 Marks)

OR

b.

i. Describe the following types of transpiration in plants in TWO points each. (4 Marks)

- Cuticular transpiration
- Stomatal transpiration

ii. Describe any THREE points of significance of transpiration in plants. (3 Marks)

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